HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Information and Communications Technology

Software Requirement Specification

An Internet Media Store

Subject: ITSS Software Development

Group 6

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1 Introduction

The following sections provide an overview of the Software Requirements Specification (SRS) for the AIMS: An Internet Media Store project. This document outlines the functional and non-functional requirements necessary for the development of the system. It serves as a reference for developers, testers, and stakeholders to ensure that the software meets its intended purpose. This document focuses on defining what the system must do, rather than its design or implementation details.

1.1 Objective

The **objective** of this document is to define the requirements of the **AIMS: An Internet Media Store** system. It serves as a formal agreement between stakeholders, developers, and testers to ensure a clear understanding of the system's capabilities. The intended audience includes:

- **Software Developers**: To implement the functionalities as specified.
- **Project Managers & Product Owners**: To validate the requirements and track progress.
- **Testers**: To design test cases based on the requirements.
- **End Users & Business Stakeholders**: To confirm that the software aligns with business needs.

This document ensures that all parties involved have a clear understanding of the system's requirements and expectations before development begins.

1.2 Scope

The **AIMS: An Internet Media Store** is a **desktop-based e-commerce application** designed for the sale and management of **physical media products**, including books, CDs, LP records, and DVDs. The software enables seamless interaction between **customers, product managers, and administrators** while ensuring performance, availability, and security standards are met.

- **Key Features:**
- **Product Management**: Product managers can **add, edit, delete, and update** media products while adhering to business rules (e.g., pricing constraints, stock limitations).
- **User Management**: Administrators can create, modify, and delete user accounts, assign roles, and reset passwords.

- **Shopping & Ordering**: Customers can search for products, view details, add them to a shopping cart, and proceed with checkout.
- **Payment Processing**: Integration with **VNPay Sandbox** for secure online transactions.
- **Order Management**: Customers can view, cancel, or modify orders before processing.
- **Delivery Options**: Standard and rush order delivery services are available, with automated shipping fee calculations.
- **Security & Logging**: The system maintains logs of product modifications and administrative actions, ensuring transparency and traceability.

- Exclusions:

- The system **does not support digital product sales**
- Payment is limited to **credit cards via VNPay**, with future expansion potential.
- Customers **do not require an account** to place an order.

This software is intended for **retail businesses**, allowing **efficient inventory management, order processing, and secure transactions**, ensuring reliability and scalability.

1.3 Glossary

No	*Term*	*Explanation*	*Example*	*Note*
1	Token	A piece of data created by the server that contains the user's information and a special authentication code. Users can pass this token to the server instead of credentials	JSON Web Token (JWT)	Compact, URL-safe, and widely used in web authentication
2	VAT	Value-Added Tax, a consumption tax applied to product prices.	10% VAT	The system calculates and adds VAT to the final price
3	API (Application Programming Protocol)	A set of rules, protocols, and tools that allows different software apps to communicate with each other.	VNPay API	AIMS connects to the API of VNPay for transactions

4	Rush Order	An expedited shipping service delivering products within 2 hours for eligible locations.	Hanoi inner-city orders	Only available in certain regions
5	VNPay	A Vietnamese online payment gateway used for processing credit card transactions in AIMS.	VNPay Sandbox	Used for secure payment processing
6	Session	A temporary period of interaction or engagement between a user and a system, during which the user accesses and interacts with software or a website, and the system maintains relevant settings and information.	software session	
7	Authentication	The process of verifying a user's identity before granting access.	Login with username and password	Uses tokens for security.
8	Credit card	A payment card issued by a financial institution, such as a bank or credit union, that allows cardholders to borrow funds up to a predetermined limit to make purchases or pay for goods and services		AIMS currently supports for credit card payment through VNPay
9	Payment gateway	A service that processes credit card payments securely.	VNPay	Handles transactions between customers and AIMS.

1.4 References

- IT4549E: Software Design and Construction - Nguyen Thi Thu Trang, trangntt@soict.hust.edu.vn.

2 Overall Description

2.1 Survey

The system in question is an online media store designed to enable customers to purchase digital media products conveniently. This software functions as a comprehensive platform, catering not only to customers but also to product managers.

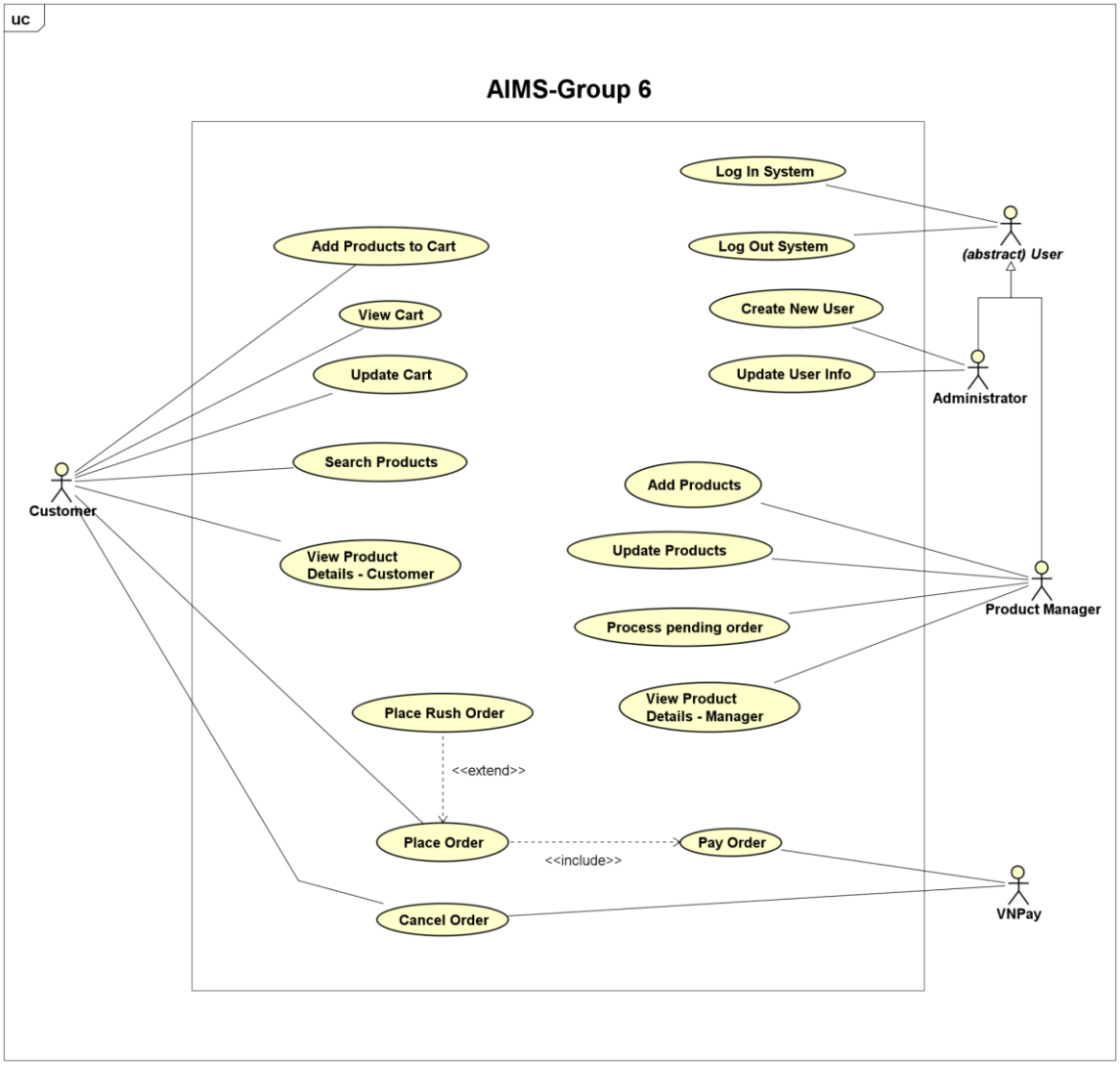
The system consists of three primary actors:

- **Customer**: Customers can look at, search for, and sort products in the store. To buy something, they must add or update items in their cart and provide delivery details. Once all information is correctly entered, they can proceed with payment via VNPay. Upon a successful transaction, AIMS will send an email confirmation along with the invoice. In certain cases, customers also have the option to place rush orders..
- **Product Manager**: Using the management interface provided by AIMS, product managers can efficiently oversee their store's inventory. They have full control over adding, removing, and updating product details. Additionally, they can manage pending orders by either approving or rejecting them.
- **Administrator**: Administrators hold the authority to manage users within the system. They can create new user accounts, update user information, and block or unblock users as needed.

2.2 Overall requirements

A Use Case diagram shows how actors interact with different use cases in the system. It represents the system's functional requirements by illustrating the interactions between external and internal agents.

The diagram below presents the general use case diagram of AIMS software, highlighting the actors and their associated use cases.

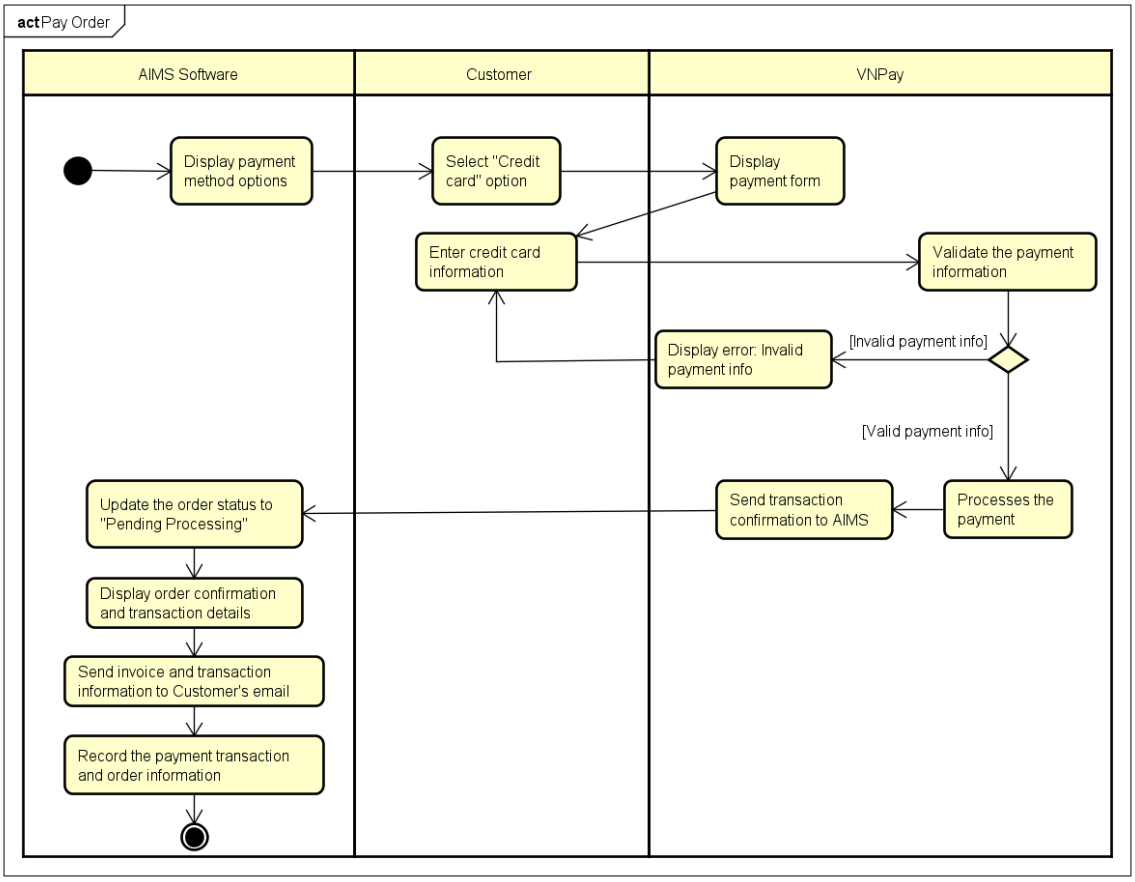


2.3 Business process

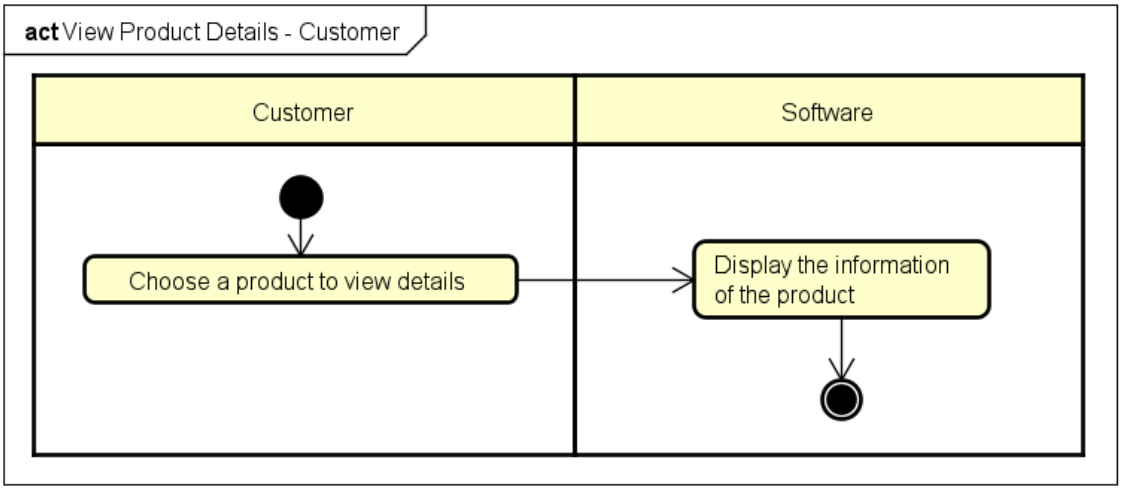
In the AIMS software, five core business operations are defined: "Place Order," "Place Rush Order," "Pay Order," "Cancel Order," and "Process Pending Order." Each of these operations is thoroughly illustrated using an activity diagram in its respective section. Below, we present the activity diagrams for all use cases in the AIMS software.

Activity diagrams are utilized to model business processes as they provide a structured visual representation of workflows, decision points, and resource allocation. These diagrams enable stakeholders to understand, analyze, and refine complex processes by depicting the sequence of activities, concurrency, error handling, and integration with other models. Additionally, they function as effective documentation tools, enhancing communication and collaboration among stakeholders.

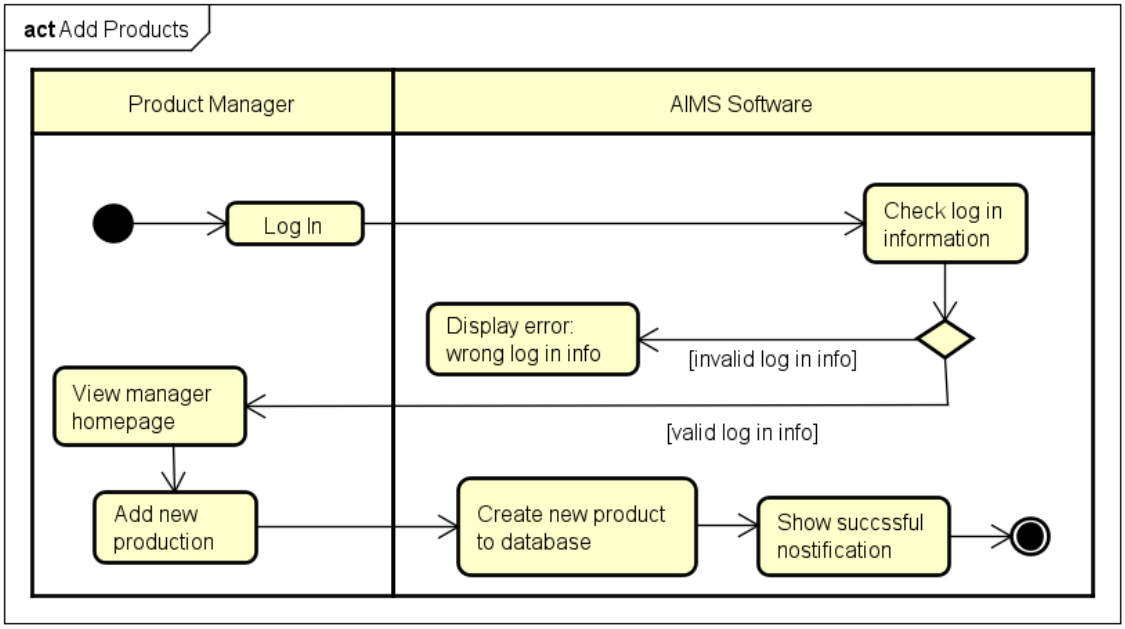
2.3.1 Business operation- Pay Order



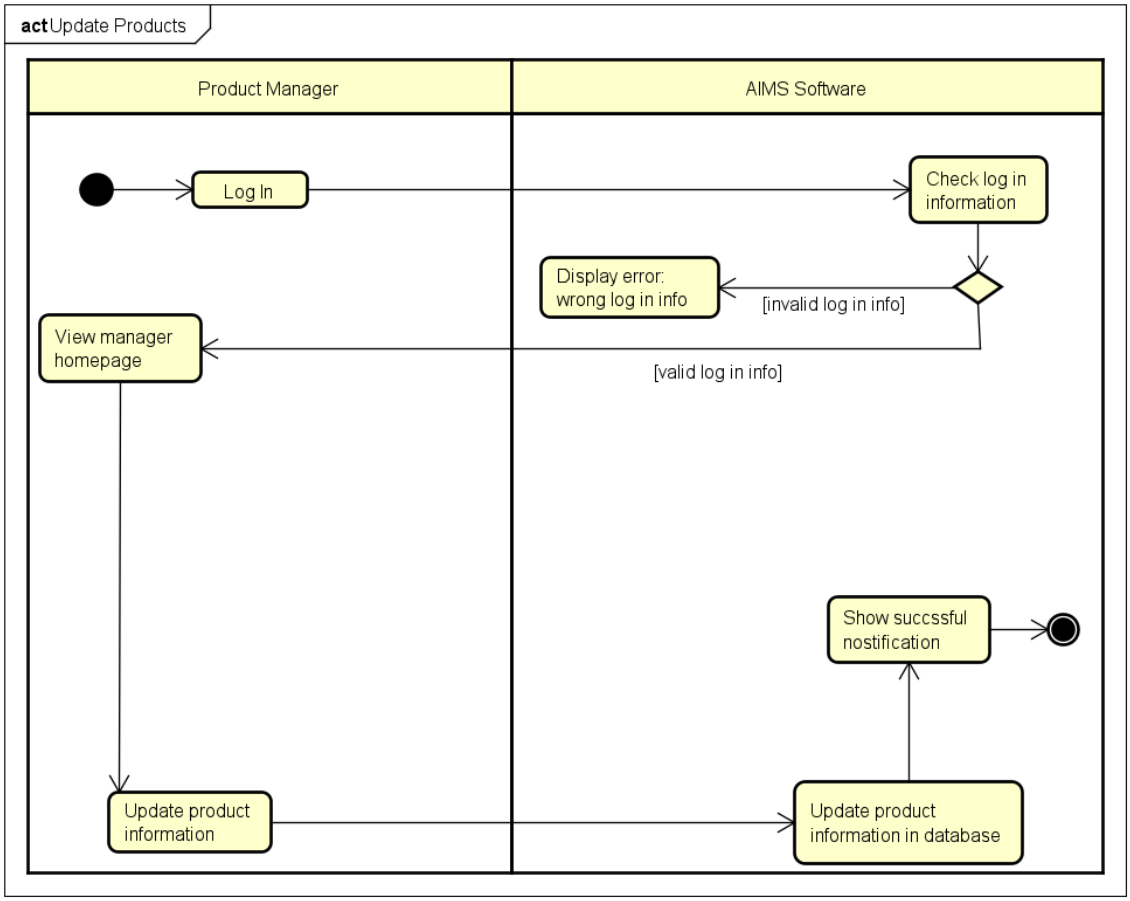
2.3.2 Business operation- View Product Detail



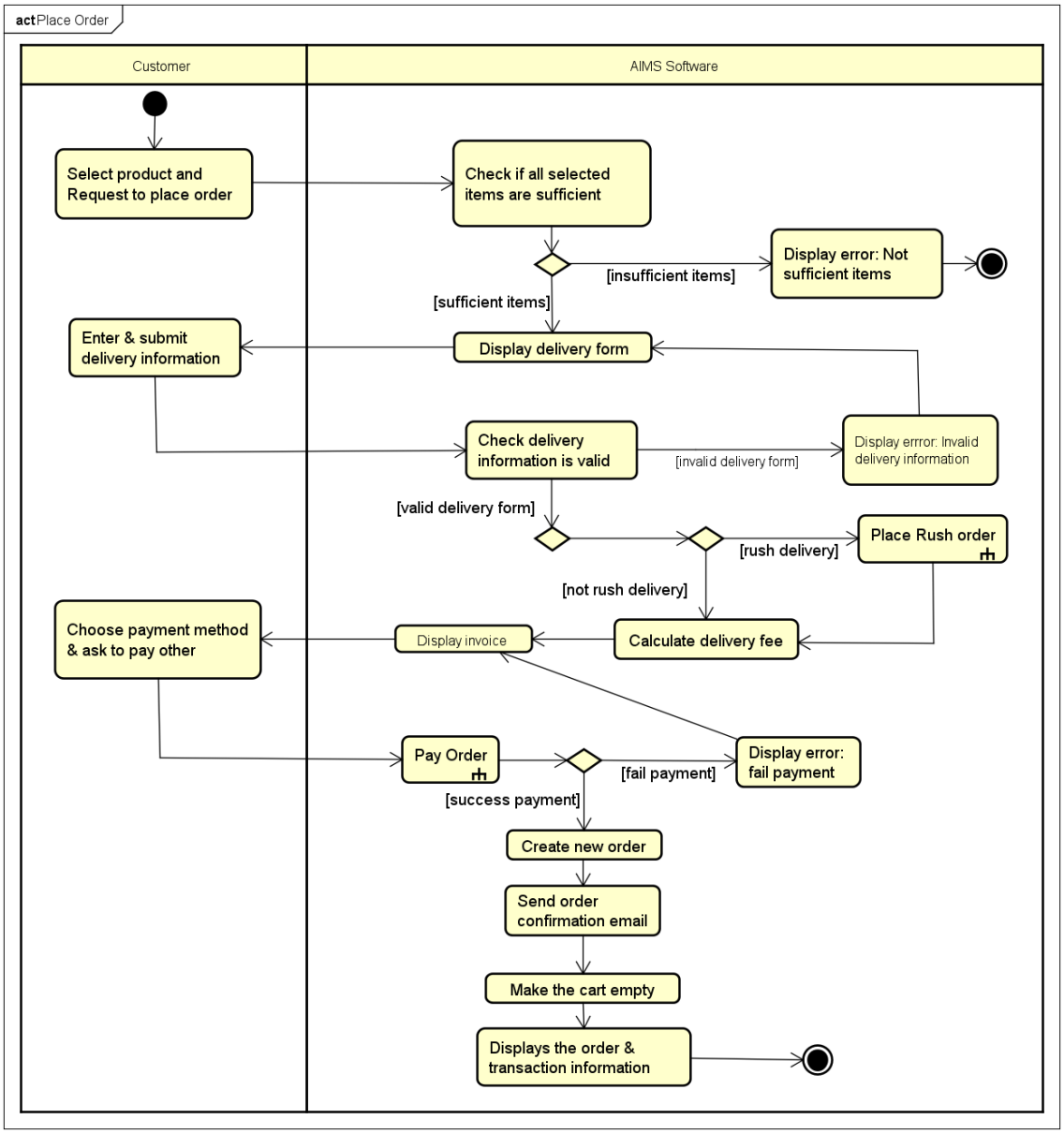
2.3.3 Business operation- Add Product



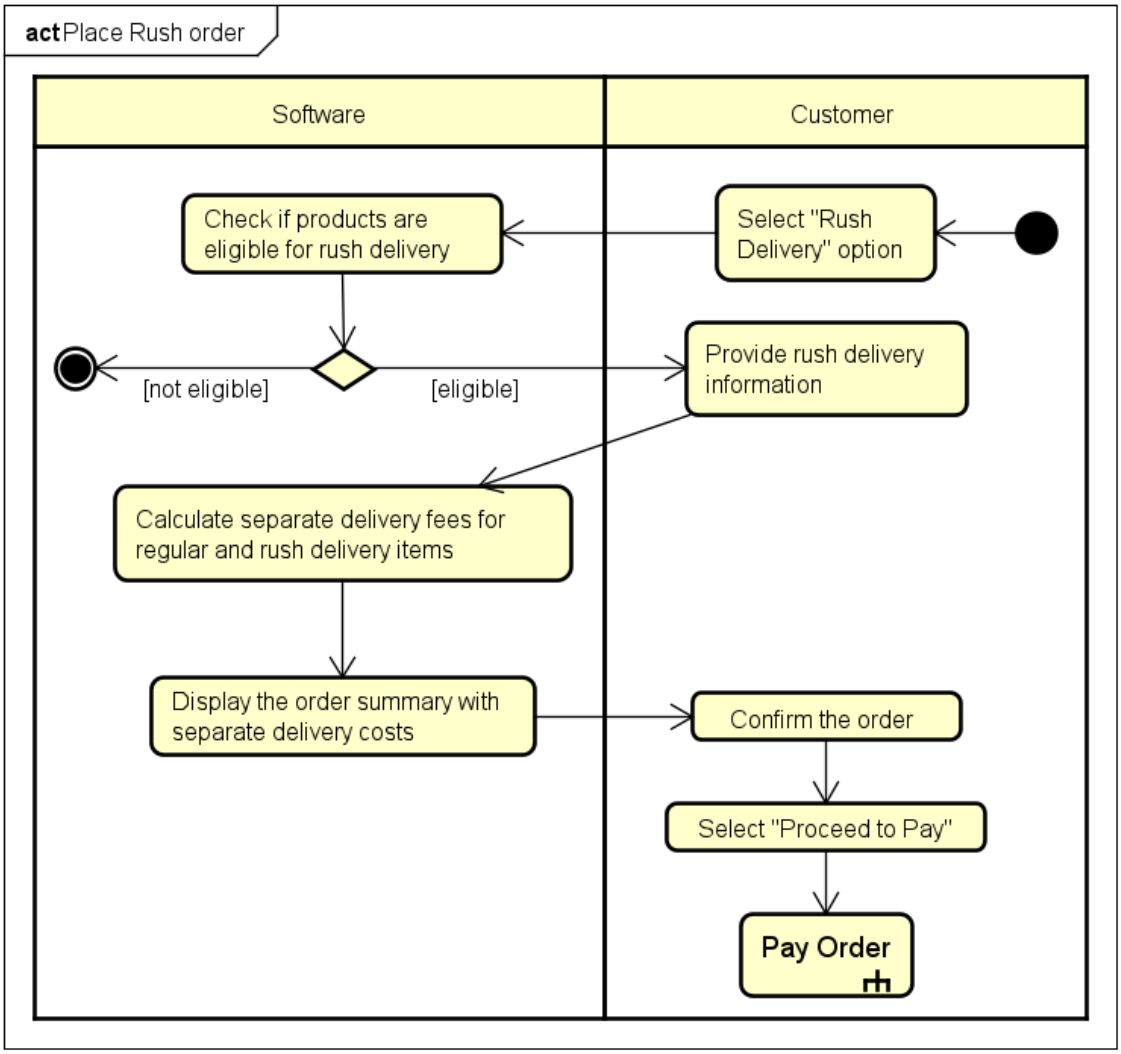
2.3.4 Business operation- Update Product



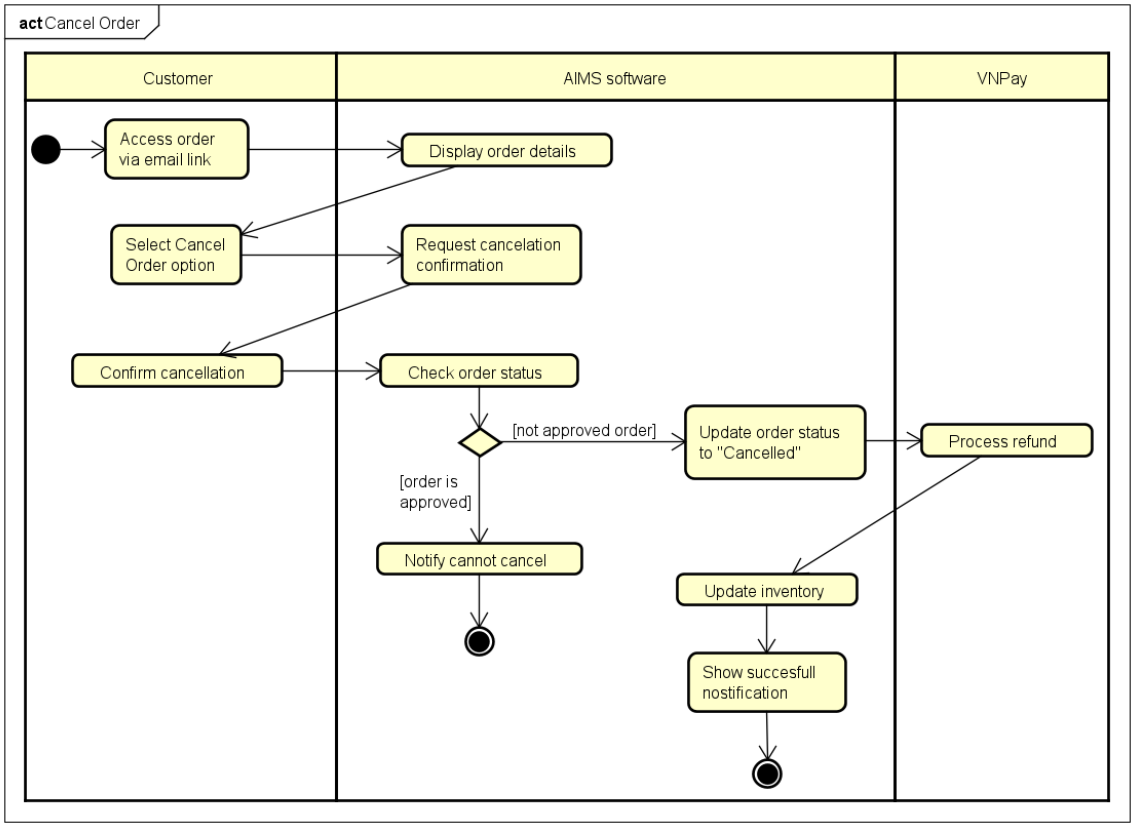
2.3.5 Business operation- Place Order



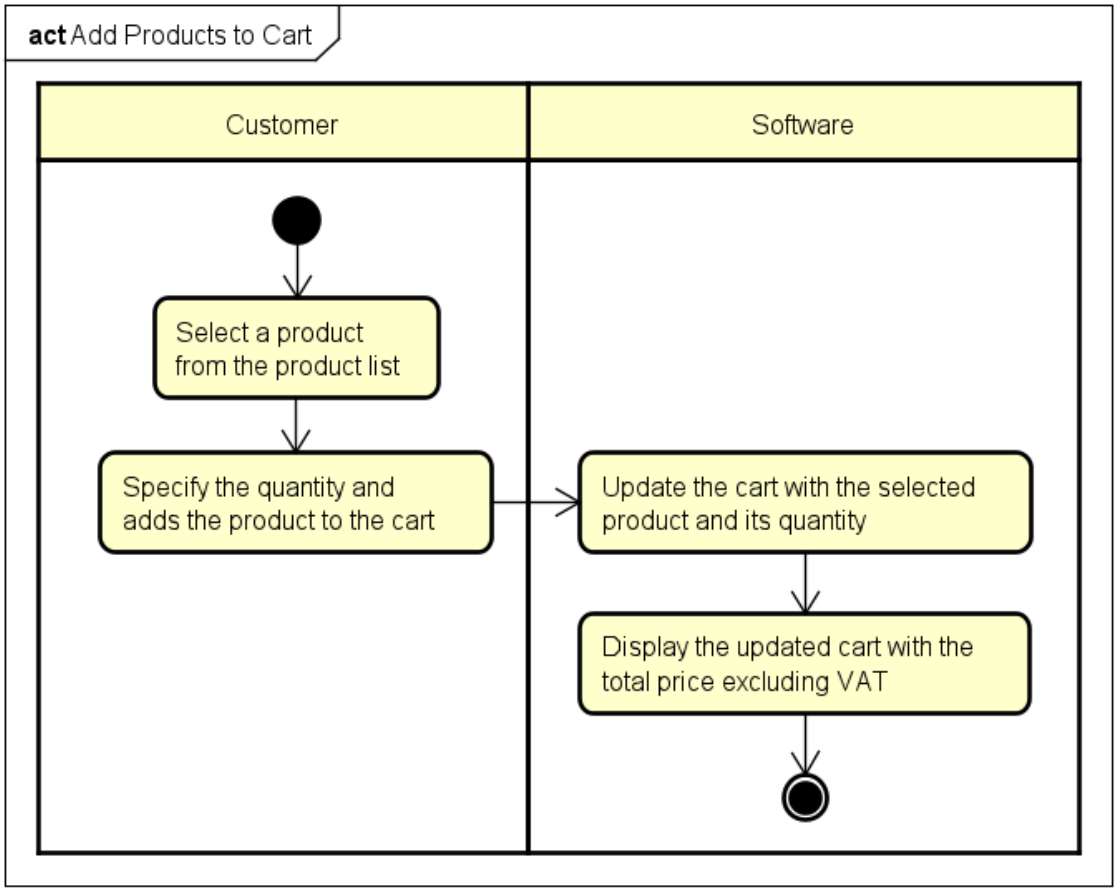
2.3.6 Business operation- Place Rush Order



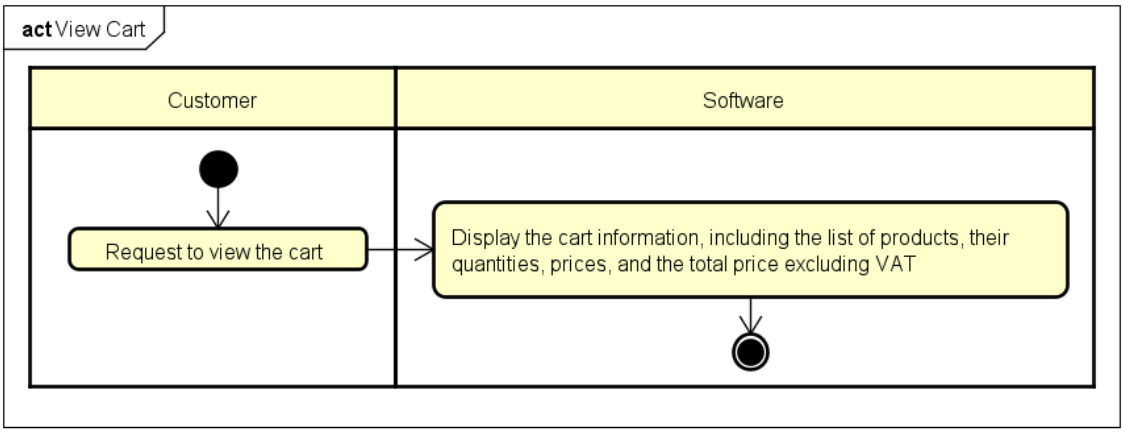
2.3.7 Business operation- Cancel Order



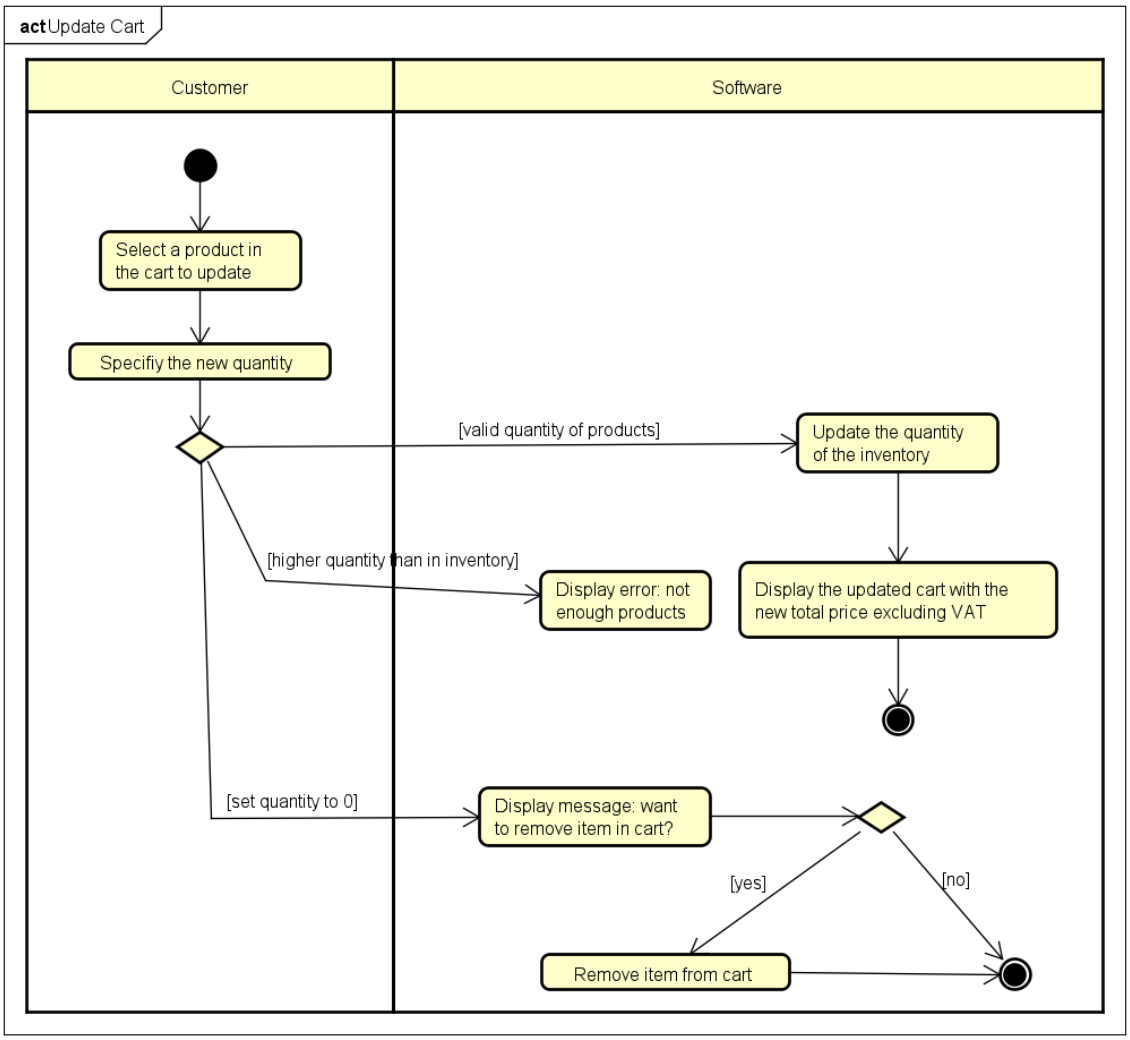
2.3.8 Business operation- Add product to Cart



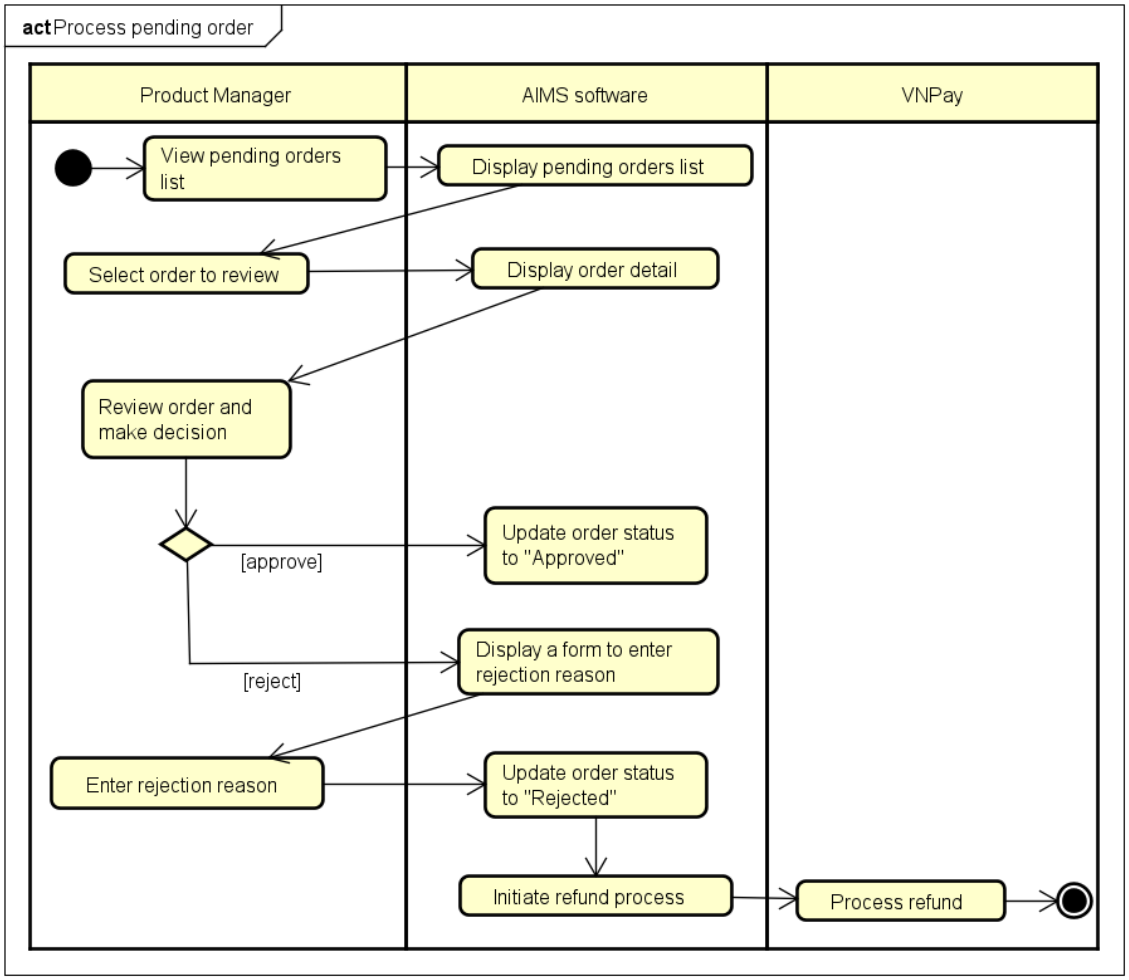
2.3.9 Business operation- View Cart



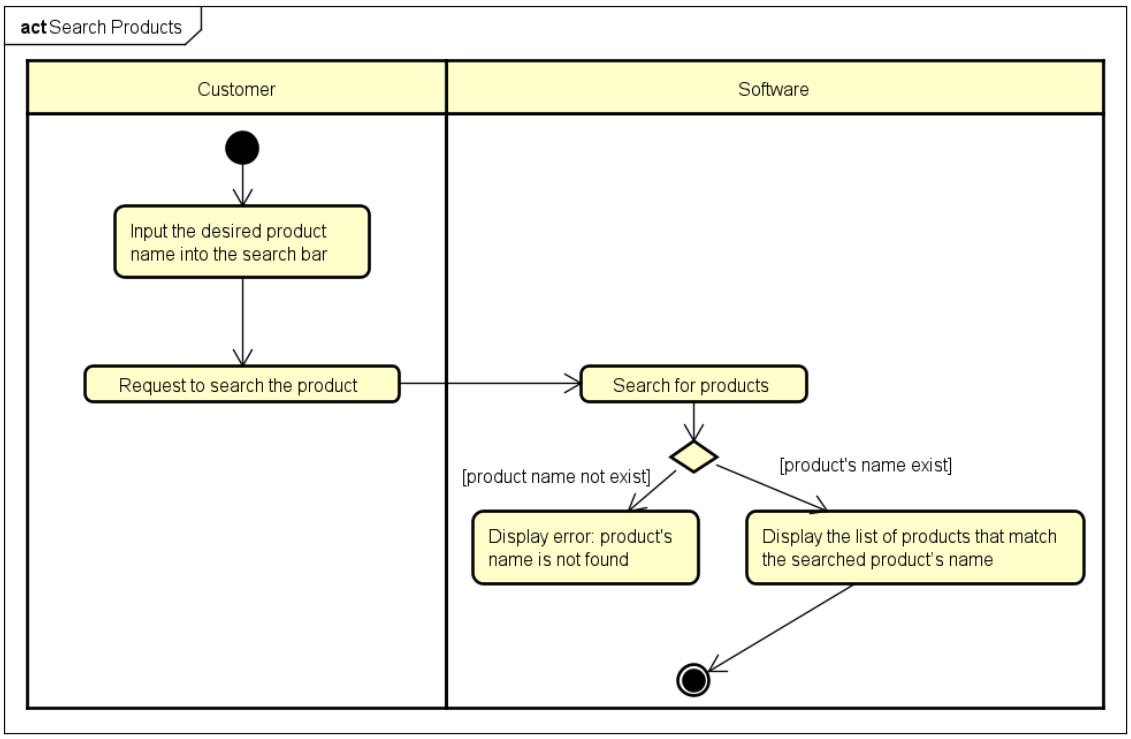
2.3.10 Business operation- Update Cart



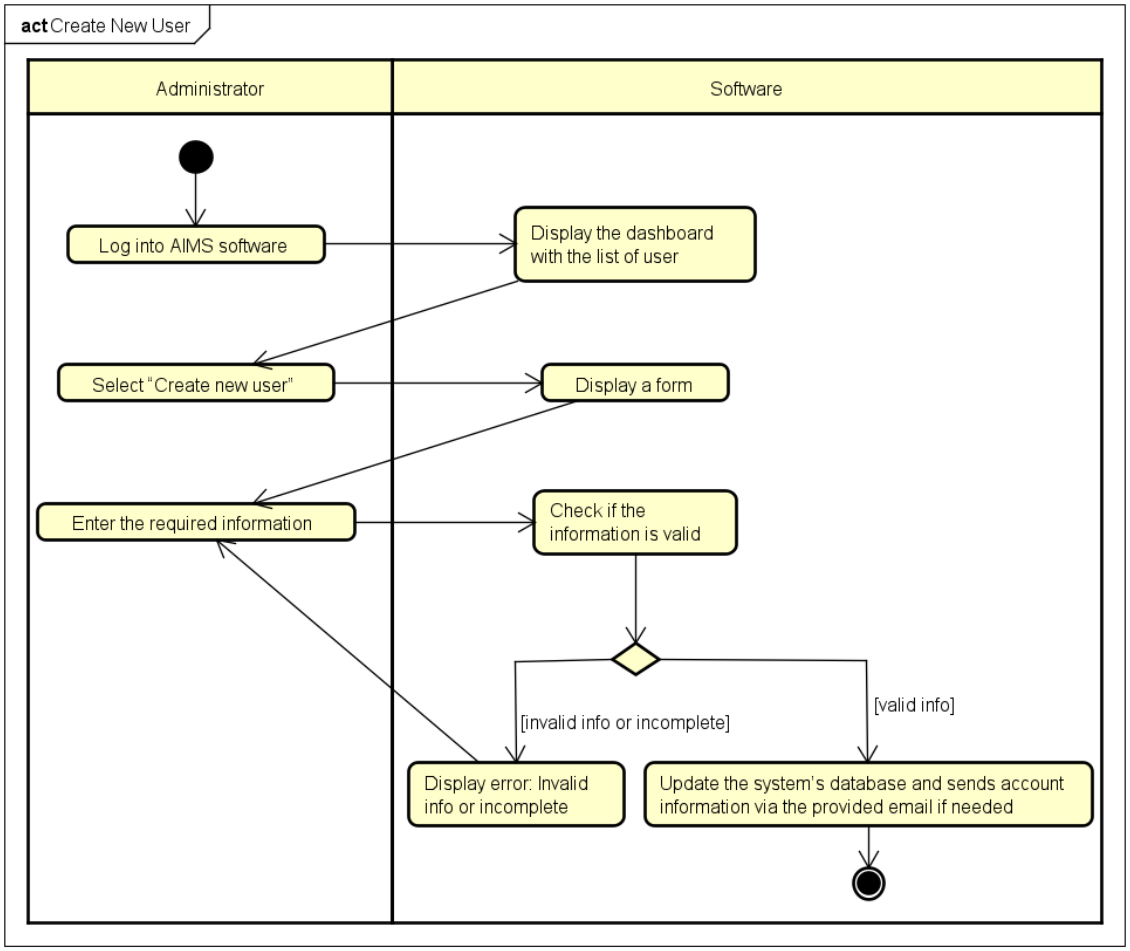
2.3.11 Business operation- Process Pending Order



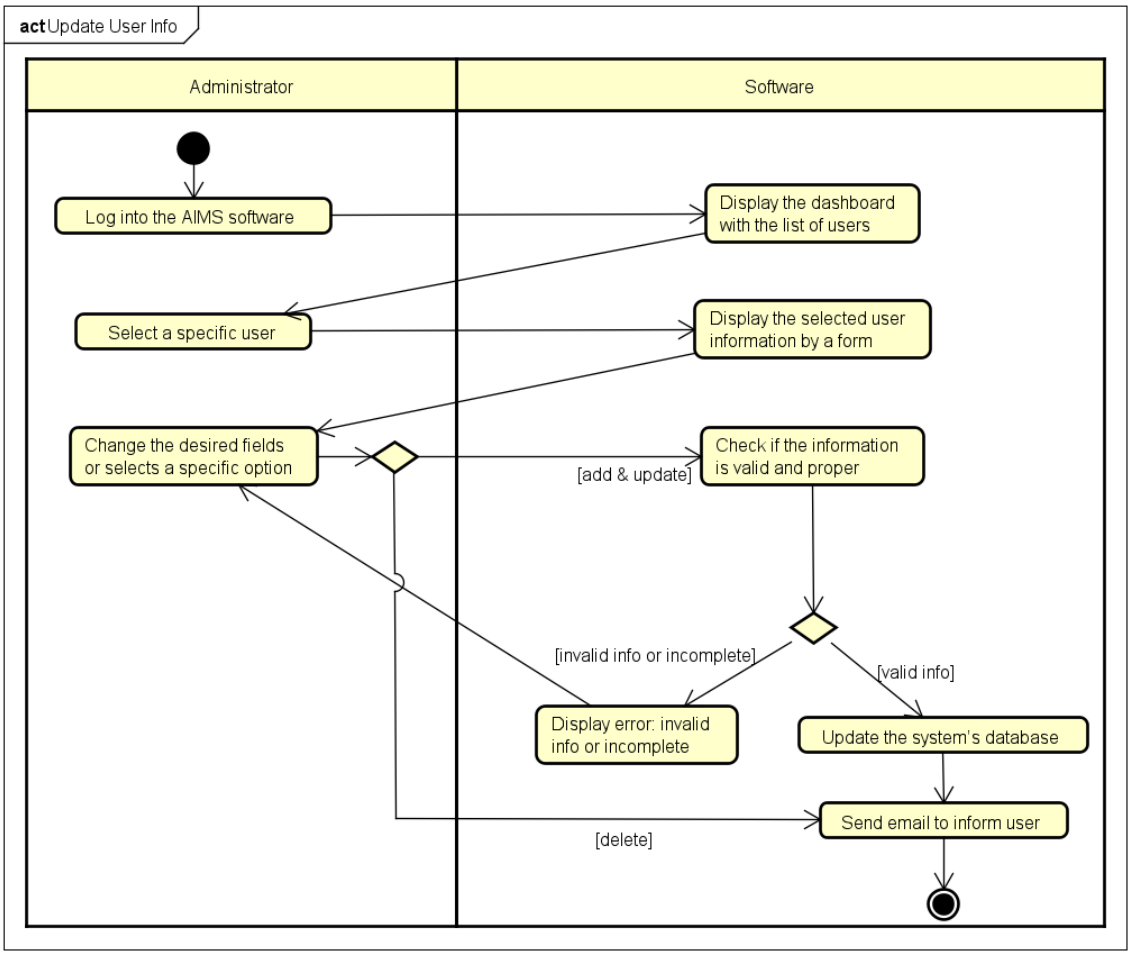
2.3.12 Business operation- Search Product



2.3.13 Business operation- Create User



2.3.14 Business operation- Update User Information



3 Detailed Requirements

Detailed requirements typically outline specific descriptions of functionalities, features, user interactions, system behavior, performance criteria, constraints, and dependencies within a software system. They serve as a core foundation for software development, covering both functional and non-functional aspects of the system.

3.1 Pay Order

Use Case "Pay Order"

1. Use case code

UC001

2. Brief Description

This use case describes the interaction between customers, software and VNPay when the customer wishes to pay an order.

3. Actors

3.1. Customer

3.2. VNPay

4. Preconditions

1. The customer submitted delivery information
2. AIMS computed the total payable amount.

5. Basic Flow of Events

1. AIMS software displays the invoice.
2. Customer asks to pay the invoice
3. AIMS software redirects to VNPay
4. VNPay sends payment result to VNPay
5. AIMS software saves invoice and payment transaction
6. VNPay notifies the payment result

6. Alternative flows

Table -Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step 5	If the customer cancels the payment transaction	• AIMS software notifies that the payment transaction is canceled	At Step 1

2.	At Step 5	If the credit card is invalid	• AIMS software notifies that the credit card is invalid to take payment transaction	At Step 1
3.	At Step 5	If the balance is insufficient	• AIMS software notifies that the balance is insufficient to complete the order.	At Step 1

7. Input data

Table A - Input data of Credit Card Information

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Card Number	Credit card number	Yes	Valid card number format	411111111111111
2.	Cardholder Name	Name on card	Yes	Non-empty string	NGUYEN VAN A
3.	Expiration Date	Card expiry date	Yes	Valid MM/YY format	12/25
4.	CVC/CVV	Card verification code	Yes	3-4 digit number	123

8. Output data

Table B - Output data of Payment Confirmation

No	Data fields	Description	Display format	Example
1.	Order Summary	General order information	Text	Customer: Nguyen Van A Phone: 0912345678 Address: 123 Tran Phu, Hanoi Total: 132,000 VND
2.	Transaction ID	Unique payment identifier	Alphanumeric	CD "Greatest Hits", 1, 150,000 VND

3.	Transaction Content	Payment description	Text	250,000 VND
4.	Transaction Datetime	Time of payment	Date/Time	15/02/2023 14:30:25

9. Postconditions

1. Order is created and saved with status "Pending Processing"
2. Inventory is updated to reserve ordered products
3. Payment transaction is recorded
4. Invoice and transaction details are sent to Customer's email

3.2 View Product Detail

Use Case "View Product Detail"

1. Use case code

UC002

2. Brief Description

This use case describes the interaction between customer and AIMS system when customer wishes to view detail information of products

3. Actors

3.1. Customers or Product Manager

4. Preconditions

None

5. Basic Flow of Events

1. The user chooses a product to view details
2. The AIMS software displays the information of the product

6. Alternative flows

None

7. Input data

None

8. Output data

Table A - Common information of product in all category:

No	Data fields	Description	Display format	Example
1.	Title	Name of product	String	'Star War'

2.	Price	Price of product		
3.	Category	Category of product	String	'book'
4.	Avail	Number of products available in store	Non-negative integer	5
5.	Image	Image of product	Image(.png file or .jpg file)	

Table B -Information of book:

No	Data fields	Description	Display format	Example
1.	Author	Author of the book	String	Vu Hai Dang
2.	Publisher	Publisher of the book	String	Fraternity
3.	Cover Type	Type of the cover	String	Paperpack
4.	Number of Page	Number of pages	Positive integer	5
5.	Publication Date	Date when book is publicated	Date (Year-Month-Day)	2025-02-27
6.	Language	Language of book content	String	English
7.	Genre	Genre type of book	String	Horror

Table C -Information of DVDs:

No	Data fields	Description	Display format	Example
1.	Disc Type	Type of DVD	String	HD-DVD
2.	Director	Director of DVD	String	Steven Sqielberg
3.	Studio	Studio publishing the movie in the DVD	String	Walt Disney
4.	Language	Language in the move	String	English
5.	Subtitle	Subtitle language	String	Germany
6.	Release date		Date (Year-Month-Day)	2025-02-27
7.	Genre	Main genre of movie	String	Comedy

Table D - Information of CDs:

No	Data fields	Description	Display format	Example

1.	Artist	Name of Artist	String	Selena Gomez
2.	Record Label		String	Warner Music
3.	Genre	Genre in the move	String	Pop
4.	Release date		Date (Year-Month-Day)	2025-02-27
5.	Track list	List of the song in the CD	String	1, Call me weh you break up 2, Scared of losing you

.

9. Postconditions

None

3.3 Add Product

Use Case "Add Product"

1. Use case code

UC003

2. Brief Description

This use case describes the interaction between the Product Manager and AIMS when adding a new product to the store

3. Actors

3.1. Product Manager

4. Preconditions

1. The Product Manager must be logged in.
2. The Product Manager has the necessary permissions to add products.

5. Basic Flow of Events

1. The Product Manager selects the option to add a new product.
2. The AIMS software displays the product entry form.
3. The Product Manager enters product details (title, category, value, price, etc.).
4. The AIMS software validates the input fields.
5. The Product Manager submits the form.
6. The AIMS software stores the new product in the database.
7. The AIMS software confirms the successful addition of the product.
8. The system updates the product list.

6. Alternative flows

Table N-Alternative flows of events for UC Add Product

No	Location	Condition	Action	Resume location
1.	At Step 3	If any required field is missing or incorrect	• AIMS software highlights invalid fields	At Step 2

7. Input data

Table A-Input data for Adding a Product

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Title	Name of the product	Yes	Non-empty string	"Dune"
2.	Category	Product category	Yes	Selected from predefined list	Book
3.	Value	Base price of the product (before VAT)	Yes	Positive number	100,000
4.	Price	Selling price (before VAT)	Yes	Between 30%-150% of Value	120,000
5.	Barcode	Unique product identifier	Yes	Alphanumeric string	1234567890
6.	Description	Short product description	No	String	Classic novel
7.	Quantity	Number of items available in stock	Yes	Non-negative integer	50
8.	Warehouse Entry Date	Date the product was added to inventory	Yes	Valid date format (YYYY-MM-DD)	2025-02-28
9.	Dimensions	Product dimensions (L x W x H)	Yes	Numeric values in cm	20x15x5 cm
10.	Weight	Product weight	Yes	Numeric value in kg	0.5 kg

8. Output data

Table B-Output data for Adding a Product

No	Data fields	Description	Display format	Example
1.	Product name	Name of product	Text	Dune
2.	Price	Selling price (before VAT)	Currency	120,000
3.	Category	Category of product	String	'book'
4.	VAT	10% of the selling price	Currency	12,000
5.	Total price	Price including VAT	Currency	132,000
6.	Stock quantity	Available units	Integer	40

9. Postconditions

1. The newly added product is stored in the system.

3.4 Update Product

Use Case "Update Product"

1. Use case code

UC004

2. Brief Description

This use case describes the interaction between the Product Manager and AIMS when updating product details in the store

3. Actors

3.1. Product Manager

4. Preconditions

1. The Product Manager must be logged in.
2. The Product Manager has the necessary permissions to update products.

5. Basic Flow of Events

1. The Product Manager selects a product to update.
2. The AIMS software displays the product details.
3. The Product Manager edits product details (title, category, value, price, etc.).
4. The AIMS software validates the updated input fields.
5. The Product Manager submits the updates.
6. The AIMS software stores the updated product details in the database.
7. The AIMS software confirms the successful update of the product.
8. The system updates the product listing.

6. Alternative flows

Table Alternative flows of events for Updating a Product

No	Location	Condition	Action	Resume location
1.	At Step 3	If any required field is missing or incorrect	• AIMS software highlights invalid fields	At Step 2
2.	At Step 6	If more than 30 products have been added in a day	• AIMS software displays a restriction message and denies the request	Use case ends

7. Input data

Table A-Input data for Updating a Product

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Title	Updated name of the product	No	Non-empty string	"Dune 2"
2.	Price	Updated selling price	No	Between 30%-150% of Value	150,000

8. Output data

Table B-Output data for Updating a Product

No	Data fields	Description	Display format	Example
1.	Updated name	Name of product	text	Dune 2
2.	Updated Price	Selling price (before VAT)	Currency	150,000
3.	Category	Category of product	String	'book'
4.	VAT	10% of the selling price	Currency	12,000
5.	Total price	Price including VAT	Currency	132,000
6.	Stock quantity	Available units	Integer	40

9. Postconditions

1. The updated product details are stored in the system.

3.5 Place Order

Use Case "Place Order (Customer)"

1. Use case code

UC005

2. Brief Description

This use case describes the interaction between Customer and AIMS when Customer wishes to place an order for products in their cart.

3. Actors

3.1. Customer

4. Preconditions

The cart needs not to be empty and they must be in the working session.

5. Basic Flow of Events

1. Customer selects products and requests to place order in the cart
2. The AIMS software checks the availability of products in the cart
3. The AIMS software displays the form of delivery information with order information
4. Customer enters and submits delivery information
5. Software checks the delivery information
6. The AIMS software calculates delivery fee and updates order information with shipping fees
7. The customer selects to payment method and asks to pay order
8. The AIMS software calls UC "Pay order"
9. The AIMS software creates and saves a new order
10. The AIMS software sends email about the order notification and information
11. The AIMS software makes the cart empty
12. The AIMS software displays the successful order notification, the order and the transaction information.

6. Alternative flows

Table N-Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step 2	If inventory is insufficient for any product	▪ The software displays the inventory quantity for each insufficient product ▪ The Customer is prompted to update cart quantities	Use case ends
2.	At Step 5	If delivery information is invalid or incomplete	▪ The software highlights invalid fields	Resumes at Step 3

			▪ The Customer is prompted to correct information	
3.	At Any Step	If Customer selects "Back"	▪ The system returns to the cart screen	Use case ends
4.	At Any Step	If Customer exits the software	▪ The ordering process is terminated	Use case ends
5	At Step 6	If the user chooses to place a rush order	● AIMS software inserts use case “Place rush order”	At Step 7

7. Input data

Table A-Input data of Delivery Information

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Recipient name	Full name of recipient	Yes	Non-empty string	Nguyen Van A
2.	Email	Contact email	Yes	Valid email format	example@mail.com
3.	Phone number	Contact phone number	Yes	Valid phone number ?	0912345678
4.	Province/City	Delivery province/city	Yes	From provided list	Hanoi
5.	District	Delivery district			Hoan Kiem
6.	Delivery address	Detailed address	Yes	Non-empty text	123 Tran Phu Street
7.	Delivery Method	Regular or Rush	Yes	"Regular" or "Rush"	Regular

8. Output data

Table B-Output data of Order Summary

No	Data fields	Description	Display format	Example
1.	Product list	List of products	Table with columns: Name, Quantity, Unit Price	Book "The Art of War", 1, 100,000 VND
2.	Subtotal	Sum of all product prices excluding VAT	Currency	100,000 VND
3.	VAT	10% of subtotal	Currency	10,000 VND

4.	Total Products with VAT	Subtotal + VAT	Currency	110,000 VND
5.	Delivery Fee	Calculated based on weight and location	Currency	22,000 VND
6.	Total amount	Final amount to be paid (Total Products with VAT + Delivery Fee)	Currency	132,000 VND

9. Postconditions

1. Order information is temporarily stored
2. The system proceeds to the payment process.

3.6 Place Rush Order

Use Case "Place Rush Order "

1. Use case code

UC006

2. Brief Description

This use case describes the interaction between Customer and AIMS when Customer wishes to place a rush order for delivery within 2 hours to an eligible location.

3. Actors

3.1. Customer

4. Preconditions

Customer has added products to the cart

5. Basic Flow of Events

1. The Customer selects "Rush Delivery" option
2. The software checks if products are eligible for rush delivery
3. The Customer provides rush delivery information
4. The software calculates separate delivery fees for regular and rush delivery items
5. The software displays the order summary with separate delivery costs
6. The Customer confirms the order
7. The Customer selects "Proceed to Pay"
8. The use case ends, transitioning to "Pay Order" use case

6. Alternative flows

Table Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step 6	If delivery address doesn't support rush delivery	▪ The software notifies Customer ▪ Customer is prompted to choose regular delivery or update address	Resumes at Step 4
2.	At Step 7	If no products are eligible for rush delivery	▪ The software notifies Customer ▪ Customer is prompted to choose regular delivery	Resumes at Step 5
3.	At Step 7	If only some products are eligible for rush delivery	▪ The software displays which products are eligible ▪ Customer can choose to proceed with mixed delivery or update selections	Resumes at Step 8
4.	At Any Step	If Customer selects "Back"	▪ The system returns to the cart screen	Use case ends
5.	At Any Step	If Customer exits the software	▪ The ordering process is terminated	Use case ends

7. Input data

Table A-Input data of Delivery Information

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Recipient name	Full name of recipient	Yes	Non-empty string	Nguyen Van A
2.	Email	Contact email	Yes	Valid email format	example@mail.com
3.	Phone number	Contact phone number	Yes	Valid phone number ?	0912345678

4.	Province/City	Delivery province/city	Yes	From provided list	Hanoi
5.	District	Delivery district	Yes	Must be inner Hanoi district for rush delivery	Hoan Kiem
6.	Delivery address	Detailed address	Yes	Non-empty text	123 Tran Phu Street
7.	Delivery Method	Regular or Rush	Yes	"Regular" or "Rush"	Regular

Table B-Input data of Rush Delivery Information

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Delivery Time	Requested 2-hour window	Yes	Valid time within business hours	14:00-16:00
2.	Delivery Instructions	Special instructions	No	String	Call upon arrival

8. Output data

Table B-Output data of Order Summary

No	Data fields	Description	Display format	Example
1.	Regular Products	List of regular delivery products	Table	Book "The Art of War", 1, 100,000 VND
2.	Rush Products	List of rush delivery products	Table	CD "Greatest Hits", 1, 150,000 VND
3.	Subtotal	Sum of all product prices excluding VAT	Currency	250,000 VND
4.	VAT	10% of subtotal	Currency	25,000 VND
5.	Total Products with VAT	Subtotal + VAT	Currency	275,000 VND
6.	Regular Delivery Fee	Fee for regular items	Currency	22,000 VND
7.	Rush Delivery Fee	Fee for rush items	Currency	32,000 VND
8.	Total amount	Final amount to be paid (Total Products with	Currency	329,000 VND

		VAT + All Delivery Fees)		

9. Postconditions

1. Rush order information is temporarily stored
2. The system proceeds to the payment process.

3.7 Cancel Order

Use Case "Cancel Order"

1. Use case code

UC007

2. Brief Description

This use case describes the interaction between Customer and AIMS when Customer wishes to cancel a paid order before it is approved.

3. Actors

3.1. Customer

3.2. VNPay

4. Preconditions

1. Order has been successfully paid
2. Order is in "Pending Processing" status
3. Customer has access to the order information via email link

5. Basic Flow of Events

1. The Customer accesses order information via the link sent in the email
2. The software displays order details
3. The Customer selects "Cancel Order" option
4. The software asks for confirmation
5. The Customer confirms cancellation
6. The software initiates refund through VNPay
7. VNPay processes the refund to the original payment method
8. VNPay sends refund confirmation to AIMS
9. The software updates the order status to "Cancelled"
10. The software updates inventory to return reserved products
11. The software sends cancellation and refund confirmation email to Customer

6. Alternative flows

Table Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step 2	If order is already approved or processed	▪ The software displays message that cancellation is not possible ▪ Customer is informed to contact support	Use case ends
2.	At Step 6-8	If connection to VNPay fails	▪ The software displays error message ▪ System logs the cancellation request for manual processing< ▪ Customer is informed that the refund will be processed manually	Use case ends
3.	At Step 7	If refund is declined by VNPay	▪ The software displays error message ▪ System logs the refund failure for manual processing ▪ Customer is informed to contact support	Use case ends

7. Input data

Table A-Input data of Cancellation Confirmation

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Cancellation Confirmation	User confirmation of cancellation	Yes	"Confirm" button click	Confirm

8. Output data

Table B-Output data of Cancellation Confirmation

No	Data fields	Description	Display format	Example
1.	Order ID	Cancelled order identifier	Alphanumeric	ORD123456

2.	Refund Amount	Amount refunded	Currency	132,000 VND
3.	Refund Transaction ID	Unique refund identifier	Alphanumeric	RFD13579246
4.	Refund Datetime	Time of refund	Date/Time	15/02/2023 15:45:32

9. Postconditions

1. Order status is updated to "Cancelled"
2. Inventory is updated to return reserved products
3. Refund transaction is recorded
4. Cancellation and refund confirmation are sent to Customer's email

3.8 Add product to Cart

Use Case "Add product to Cart"

1. Use case code

UC008

2. Brief Description

This use case describes the interaction between customer and the AIMS system when customer wishes to add products to the cart.

3. Actors

3.1. Customer

4. Preconditions

The customer has navigated to the product list or product detail screen.

5. Basic Flow of Events

1. The customer selects a product from the product list.
2. The customer specifies the quantity and adds the product to the cart.
3. The software updates the cart with the selected product and its quantity.
4. The software displays the updated cart with the total price excluding VAT.

6. Alternative flows

Table N-Alternative flows of events for UC Add product to Cart

No	Location	Condition	Action	Resume location

1.	Step 3	Customer attempts to add a product quantity that exceeds available inventory	▪ Show a notification about the maximum available quantity. The customer may modify the quantity or abort the action.	Step 4

7. Input data

Table A-Input data of quantity of product.

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Quantity	The customer selects a product and specifies the quantity to update	Yes	Positive integer	10

8. Output data

None

9. Postconditions

A new product along with its quantity is added to the cart.

3.9 View Cart

Use Case "View Cart"

1. Use case code

UC009

2. Brief Description

This use case describes the interaction between customer and the AIMS system when customer wishes to view the cart.

3. Actors

3.1. Customer

4. Preconditions

The customer has navigated to the product list or product detail screen.

5. Basic Flow of Events

1. The customer requests to view the cart.
2. The software displays the cart information, including the list of products, their quantities, prices, and the total price excluding VAT.

6. Alternative flows

None

7. Input data

None

8. Output data

None

9. Postconditions

The cart is updated with the new list of products and quantities.

3.10 Update Cart

Use Case "Update Cart"

1. Use case code

UC010

2. Brief Description

This use case describes the interaction between customer and the AIMS system when customer wishes to update the cart.

3. Actors

3.1. Customer

4. Preconditions

The customer has navigated to the cart screen.

5. Basic Flow of Events

1. The customer selects a product in the cart to update.
2. The customer specifies the new quantity.
3. The software updates the quantity of the inventory.
4. The software displays the updated cart with the new total price excluding VAT.

6. Alternative flows

Table Alternative flows of events for UC Update Cart

No	Location	Condition	Action	Resume location
1.	Step 2	Customer attempts to update a product quantity that exceeds	▪ Display a message indicating the maximum available quantity. Customer can adjust the quantity or cancel the action.	Use case ends

		available inventory		
2.	Step 4	Customer set the quantity to 0	▪ Display a confirmation message to remove the item from the cart.	Step 4

7. Input data

Table A-Input data of UC Update Cart

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Quantity	The customer selects a product and specifies the quantity to update	No	Non negative integer	10

8. Output data

None

9. Postconditions

The cart is updated to reflect the adjusted quantity of the modified items or the removal of the selected product.

3.11 Process Pending Order

Use Case "Process Pending Order"

1. Use case code

UC011

2. Brief Description

This use case describes the interaction between Product Manager and AIMS when Product Manager reviews and approves or rejects pending orders.

3. Actors

3.1. Product Manager

3.2. VNPay

4. Preconditions

1. Product Manager is logged into the system
2. One or more orders are in "Pending Processing" status

5. Basic Flow of Events

1. The Product Manager accesses the list of pending orders

2. The software displays up to 30 pending orders per page
3. The Product Manager selects an order to review
4. The software displays detailed order information
5. The software validates current inventory availability
6. The Product Manager reviews the order
7. The Product Manager decides to approve the order
8. The software updates the order status to "Approved"
9. The software sends order approval notification to Customer's email

6. Alternative flows

Table N-Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step 5	If inventory is insufficient	▪ The software automatically displays insufficient inventory warning ▪ The software prevents approval	Use case ends
2.	At Step 7	If Product Manager decides to reject the order	▪ Product Manager enters rejection reason ▪ The software updates order status to "Rejected" ▪ The software initiates refund through VNPay ▪ The software sends rejection notification to Customer's email	Use case ends
3.	At Step 8	If VNPay refund connection fails (after rejection)	▪ The software displays error message ▪ System logs the refund request for manual processing ▪ Product Manager is informed that the refund will be processed manually	Use case ends

7. Input data

Table A-Input data of Process Pending Order

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Rejection Reason	Reason for rejecting the order	Yes	Non-empty string	Out of stock while customer was paying

8. Output data

Table B-Output data of Process Pending Order

No	Data fields	Description	Display format	Example
1.	Order ID	Order identifier	Alphanumeric	ORD123456
2.	Customer Info	Customer details	Text	Nguyen Van A 0912345678 example@mail.com
3.	Order Contents	List of products	Table	Book "The Art of War", 1, 100,000 VND
4.	Payment Info	Payment details	Text	Paid via Credit Card VNP13579246 15/02/2023 14:30:25
5.	Delivery Info	Delivery details	Text	123 Tran Phu, Hoan Kiem, Hanoi
6.	Order Status	Current order status	Text	Pending Processing

9. Postconditions

1. Order status is updated to either "Approved" or "Rejected"
2. If rejected, refund is processed and customer is notified
3. If approved, customer is notified of order approval

3.12 Search Product

Use Case "Search Product"

1. Use case code

UC012

2. Brief Description

This use case describes the interaction between customer and the AIMS system when customer wishes to search for a product.

3. Actors

3.1. Customer

4. Preconditions

None

5. Basic Flow of Events

1. Customer inputs the desired product name into the search bar.
2. Customer request to search the product.
3. AIMS searches for products.
4. AIMS displays the list of products that match the searched product's name.

6. Alternative flows

Table N-Alternative flows of events for UC Search Product

No	Location	Condition	Action	Resume location
1	Step 4	If no product matches the given name.	▪ AIMS notifies that no product matches the given name	Step 4

7. Input data

Table A-Input data of Search Product

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Title of product	Name of product to search	No	String	Marvel

8. Output data

Table B -Output data of UC Search Product:

No	Data fields	Description	Display format	Example
1.	Title	Name of product	String	'Star War'
2.	Price	Price of product		
3.	Category	Category of product	String	'book'
4.	Avail	Number of products available in store	Non-negative integer	5
5.	Image	Image of product	Image(.png file or .jpg file)	

9. Postconditions

None

3.13 Create New User

Use Case "Create New User"

1. Use case code

UC013

2. Brief Description

This use case describes the interaction between Administrator and AIMS when Administrator creates a new User and provides mandatory information.

3. Actors

3.1. Administrator

4. Preconditions

1. Administrator is logged into the system

5. Basic Flow of Events

5.1. Administrator log into AIMS software

5.2. The AIMS software displays the dashboard with the list of users.

5.3. Administrator select "Create new user".

5.4. The AIMS software displays a form.

5.5. Administrator enters the required information.

5.6. The AIMS software checks if the information is valid.

5.7. The AIMS software updates the system's database and sends account information via the provided email if needed.

6. Alternative flows

Table N-Alternative flows of events for UC Create New User

No	Location	Condition	Action	Resume location
1.	At Step 4	If the entered information is invalid or incomplete	▪ The AIMS software asks you to enter errored fields.	Step 2.

7. Input data

Table A-Input data of Create New User

No	Data fields	Description	Mandatory	Valid condition	Example
1	Account name	Nothing specific	Yes	Non-empty string No duplicate with available account names	Out of stock while customer was paying
2.	Password	Can enter hiddenly	Yes	At least 8 characters	12345678
3.	Role	Select from a list	Yes	Administrator/Product Manager	Administrator
4.	Email		Yes	Existent	hieu.hq210111@@sis.hust.edu.vn

8. Output data

None

9. Postconditions

1. The managed list of users is updated and is displayed on screen.

3.14 Update User Information

Use Case "Update User Information"

1. Use case code

UC014

2. Brief Description

This use case describes the interaction between Administrator and AIMS when Administrator updates a selected user (e.g account information, update password, modify role).

3. Actors

3.1. Administrator

4. Preconditions

1. Administrator is logged into the system

5. Basic Flow of Events

5.1. Administrator logs into the AIMS software.
5.2. The AIMS software displays the dashboard with the list of users.
5.3. Administrator selects a specific user.
5.4. The AIMS software displays the selected user information by a form.
5.5. Administrator changes the desired fields or selects a specific option.
5.6. The AIMS software checks if the information is valid and proper.
5.7. The AIMS software updates the system's database
5.8. The AIMS software sends account information via the provided email.

6. Alternative flows

Table N-Alternative flows of events for UC Update User Information

No	Location	Condition	Action	Resume location
1.	At Step 6	If the entered information is invalid or incomplete	▪ The AIMS software asks you to enter errored fields.	Step 4.
2.	At Step 4	If Administrator deletes the selected user	▪ The AIMS software notifies the user from their email. ▪ The AIMS software returns to the dashboard.	Step 2

7. Input data

Table A-Input data of Update User Information

No	Data fields	Description	Mandatory	Valid condition	Example
1	Account name	Nothing specific	No	Non-empty string No duplicate with available account names	Out of stock while customer was paying
2.	Password	Can enter hiddenly	No	At least 8 characters	12345678
3.	Role	Select from a list	No	Administrator/Product Manager	Administrator

4.	Email	User's available email account	No	Existent	hieu.hq210111@ @sis.hust.edu.vn
5.	State	Toggle a button	No	Blocked/Unblocked	Blocked

8. Output data

None

9. Postconditions

1. AIMS returns to the dashboard.
2. The managed list of users is updated and is displayed on screen.

4 Supplementary specification

4.1 Functionality

For use cases involving transactional operations, if errors occur during the connection or execution process, the system should provide appropriate error notifications to clarify that the issue is system-related rather than user-induced.

For use cases involving registration, account recovery, and password changes, the system must ensure accurate, secure, and reliable confirmation, utilizing at least two verification methods to authenticate the user's account. Situations where insufficient information prevents account verification should be avoided.

Use cases accessible by Product Managers, Customers, and Guest Users require login with the appropriate roles.

4.2 Usability

- Functions should be designed for intuitive and efficient operation.
- The layout should be user-friendly and easy to navigate.
- The language used should be clear and easily understandable, avoiding excessive technical jargon.

4.3 Reliability

Availability: The system should remain operational the majority of the time, with scheduled maintenance restricted to a maximum of one hour per month.

Error Handling: A robust error management system should detect and log errors while notifying administrators for prompt resolution. The system must handle errors gracefully to minimize service disruptions and ensure a seamless user experience.

Testing and Verification: Comprehensive testing, including unit tests, integration tests, and stress tests, should be conducted to validate the system's reliability under various conditions. Automated monitoring tools should continuously track performance and notify administrators of potential issues.

Documentation and Reporting: Detailed documentation should cover system architecture, failover procedures, and incident response protocols. Reliability incidents and resolutions should be recorded in incident reports to support continuous improvement and accountability.

4.4 Performance

The system shall meet the following performance criteria to ensure a smooth and responsive user experience for customers, vendors and administrators.

Response time: The maximum response time of the software is 2 seconds under normal conditions or 5 seconds during peak hours.

Throughput: This software can serve up to 1,000 customers simultaneously without significantly reducing performance and can operate continuously for 300 hours without failure.

Recovery: Additionally, the software can resume normal operation within a maximum of 1 hour after an incident

4.5 Supportability

The system shall be designed for long-term maintainability and ease of support, ensuring that it remains functional, scalable and secure with minimal downtime.

Scalability & Extensibility:

- The platform architecture shall support **modular development** to allow independent feature enhancements without disrupting core functionality.
- The system shall support **plugin-based extensibility** to add third-party integrations (e.g., payment gateways).

Customer support & Issue resolution:

- A knowledge base or FAQ section shall be available to help customers resolve common issues independently.

4.6 Other requirements

A visually appealing and user-friendly graphical interface with fast processing speed and high accuracy. Well-organized product categorization for easy navigation and efficient search.